ModelIt!

Teacher's Guide

When Trees Become Matches

California's Burning Season and the Earth Systems That Fuel It

Quick Reference

Grade Level	Grade 5
NGSS Standard	5-ESS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-ESS2-1

Develop a model using an example to describe ways the geosphere, biosphere, hydrosphere, and/or atmosphere interact.

Three-Dimensional Learning

Science & Engineering Practice	**Developing and Using Models** — Students build a computational model to represent the California wildfire system, identifying components and their relationships.
Disciplinary Core Idea	**ESS2.A: Earth Materials and Systems** — Earth's major systems interact in complex ways. The fire system demonstrates interactions between hydrosphere (rainfall), atmosphere (wind), biosphere (vegetation), and geosphere (terrain).
Crosscutting Concept	**Systems and System Models** — Students learn that models can represent systems and their interactions. They analyze cause-and-effect relationships within the system.

CAST Testing Alignment

MULTIPLE CHOICE:

Which Earth system provides the fuel for wildfires? A) Hydrosphere B) Biosphere C) Atmosphere D) Geosphere [Answer: B - vegetation is part of the biosphere]

CONSTRUCTED RESPONSE:

Explain how a decrease in rainfall affects the other components of the California fire system. Use cause-and-effect language in your response.

MODEL INTERPRETATION:

Look at this diagram of a fire system. Draw an arrow showing how wind affects fire spread. Label it as positive (+) or negative (-). Explain your reasoning.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

California wildfires are driven by the interaction of multiple Earth systems. The atmosphere controls wind and rainfall. The biosphere provides fuel in the form of vegetation. The hydrosphere determines moisture levels.

The Fire Triangle

Fire needs three things: fuel (dry vegetation), oxygen (wind provides fresh air), and heat (temperature or an ignition source). Remove any one and fire cannot exist.

California's Fire Season

California's Mediterranean climate creates a natural fire cycle: wet winters grow vegetation, hot dry summers cure it into fuel. Santa Ana winds in fall provide the final ingredient.

Earth System Interactions

Wildfires show how Earth's systems interact. The atmosphere (weather), biosphere (vegetation), hydrosphere (drought), and geosphere (terrain) all play roles.

Human Factors

90% of California wildfires are human-caused. Power lines, campfires, and equipment can ignite fires during high-risk conditions.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System

Students identify Rainfall, Dry Vegetation, Wind, and Fire Spread as the key components of California's fire system.

E — Establish Relationships

Students determine that Rainfall has a NEGATIVE effect on Dry Vegetation, while Dry Vegetation and Wind have POSITIVE effects on Fire Spread.

V — Visualize & Model

Students build the wildfire model in ModelIt!, arranging components, sorting External vs Internal, and drawing directional arrows.

E — Evaluate Outcomes

Students run drought, wind, and fire season scenarios, observing how activity graphs show component changes over time.

R — Revise & Extend

Students add Temperature, Humidity, or Santa Ana Winds to make the model more realistic based on their research.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover & Introduction

2-3 min

Say: *"Today we're going to investigate a California mystery — why does our state burn at the same time every single year? We'll use a tool called ModelIt! to build a model that helps us discover the answer."*

Watch For

Build curiosity! Ask: Has anyone seen news about California fires? When do they usually happen? Accept all responses. Be sensitive to students who may have personal experience with wildfires.

Slide 2: Learning Objectives & Vocabulary

2 min

Say: *"Let's preview the key words we'll use today. A COMPONENT is a part of a system. A MODEL is a simplified version of something complex. A SIMULATION lets us test what-if scenarios."*

Watch For

Quick check: Who can give me an example of a system? Have students turn and talk. Accept all answers — globes, diagrams, dioramas are all models.

Slide 3: The Big Question

3 min

Say: *"Here's our big question: Why does California burn at the same time every year? By the end of today's lesson, you'll be able to answer this using evidence from your model."*

Watch For

Have students make predictions in their Activity Pack Pre-Assessment. Collect ideas verbally but don't correct — they'll confirm through the model.

Slide 4: Today We Will Build a Model!

2 min

Say: *"We'll follow these five steps — LEVER — to build our model like real scientists do."*

Watch For

Emphasize that modeling is a PRACTICE scientists use. Briefly explain each step but don't belabor it — students learn by doing.

Slide 5: Activity 1 — Sort Components

5-7 min

Say: *"Here are our four components. Your job is to sort them — which ones are EXTERNAL (inputs from outside, like weather) and which are INTERNAL (things happening inside the system)?"*

Watch For

THIS IS HANDS-ON TIME. Have students open ModelIt! and Activity Packs. Walk around as pairs discuss and sort. Listen for reasoning, not just answers. Common mistake: students think fire is external because it seems to come from outside.

Slide 6: Activity 2 — Connect with Arrows

7-10 min

Say: *"Now let's connect our components. When RAINFALL increases, does DRY VEGETATION increase or decrease? That's a NEGATIVE relationship — one goes up, the other goes down."*

Watch For

Model thinking aloud. Use the test: If X increases, what happens to Y? Have students explain reasoning before drawing arrows. Watch for directionality confusion.

Slide 7: Activity 3 — Run the Simulation!

8-10 min

Say: *"Now the exciting part — let's test our model! We'll run three scenarios. First, PREDICT what you think will happen. Then RUN the simulation and OBSERVE."*

Watch For

Circulate and ask probing questions: What happened when you turned rainfall off? Why do you think fire spread increased? Have students record predictions BEFORE running each scenario.

Slide 8: Discussion & Discoveries

5 min

Say: *"Let's share what we discovered. Who can tell me what happened during the fire season scenario?"*

Watch For

Facilitate class discussion. Have multiple groups share observations. Connect findings back to the Big Question. Students should feel they DISCOVERED the answer, not that you told them.

Slide 9: STEM Challenge Preview

5-10 min

Say: *"Now you'll use what you learned to solve a real engineering problem — designing firebreaks to protect a town!"*

Watch For

If time allows, begin the STEM Challenge. Can also be homework or a Day 2 activity. Emphasize that real Fire Protection Engineers do this work.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Rainfall, Wind
- **INTERNAL:** Dry Vegetation, Fire Spread

Relationships

- **Rainfall to Dry Vegetation:** NEGATIVE (-) — More rain means less dry vegetation.
- **Dry Vegetation to Fire Spread:** POSITIVE (+) — More dry vegetation means fire spreads faster.
- **Wind to Fire Spread:** POSITIVE (+) — More wind pushes fire farther and faster.

Key Discoveries

- Rainfall and dry vegetation have an inverse relationship
- Wind makes fire spread dramatically faster
- When multiple factors combine, fire behavior becomes extreme
- Earth systems are interconnected—changes in one affect all others

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Wildfires are always bad"	Fire is a natural part of many ecosystems. Some plants need fire to reproduce. The problem is when fires burn where people live.
"Climate and weather are the same thing"	Weather is daily conditions; climate is the long-term pattern. California's climate creates fire conditions, but weather determines when fires start.
"Firefighters just spray water on fires"	Modern firefighting uses science—weather models, fuel moisture data, and terrain analysis—to predict fire behavior and plan strategy.
"More components always makes a better model"	Models should be as simple as possible while still being useful. A clear 4-component model students understand is better than a complex 10-component model they cannot explain.